Replay-Aware CVRP Benchmark Report

Overview

- Condition count: 4.
- Best final transfer gap: cvrp_failure_replay_compression.
- Best CVRPLIB holdout gap: cvrp_failure_replay_compression.
- Best synthetic holdout gap: cvrp_failure_replay_compression.

Run Metadata

- run_name: run_20260513_193323_k
- started_at_local: 2026-05-13 19:33:23
- finished_at_local: 2026-05-13 19:36:43
- duration_hhmm: 00:03
- duration_seconds: 199.268
- seed_offset: 10000
- replicate_label: k
- judge_status: enabled
- judge_provider: openai
- judge_model: gpt-4.1-mini

Condition Comparison

Condition	Replay	Selection	Compression	Final CVRPLIB Gap	Final Synthetic Gap	Final Transfer Gap	Mean Accepted Novelty	Mean Complexity	Adaptation Efficiency
cvrp_no_replay	none	score_only	False	0.244084	0.005376	0.137992	0.905812	0.58	0.03293
cvrp_random_replay	random	score_only	False	0.239694	0.005376	0.135553	0.889336	0.58	-0.003996
cvrp_failure_replay	failure	score_only	False	0.237687	0.005376	0.134438	0.880282	0.58	0.061869
cvrp_failure_replay_compression	failure	novelty_gate	True	0.198494	0.0	0.110274	0.766067	0.78	0.035528

Condition Notes

cvrp_no_replay

- Replay mode: none with selection mode score_only.
- Final transfer gaps: CVRPLIB 0.244084, synthetic 0.005376, combined 0.137992.

- Accepted-epoch count 2, mean accepted code novelty 0.905812, and final complexity 0.58.
- Adaptation efficiency 0.03293 and archive sizes {'worst_cases': 5, 'failure_cases': 4, 'adversarial_layouts': 4} / elite 1.
- Panel heldout_cvrplib mean gap 0.244084 across 5 instances; family means: A=0.099476, B=0.249482, E=0.320537, P=0.301444.
- Panel synthetic_holdout mean gap 0.005376 across 4 instances; family means: alternating_belt=0.021505, clustered_demand=0.0, corridor_split=0.0, radial_heavy=0.0.
- Worst recent training cases: [{"best_known_cost": 458, "cost": 604, "family": "P", "name": "P-n40-k5", "optimality_gap": 0.318777}, {"best_known_cost": 937, "cost": 1065, "family": "A", "name": "A-n44-k6", "optimality_gap": 0.136606}, {"best_known_cost": 784, "cost": 860, "family": "A", "name": "A-n32-k5", "optimality_gap": 0.096939}]

cvrp_random_replay

- Replay mode: random with selection mode score_only.
- Final transfer gaps: CVRPLIB 0.239694, synthetic 0.005376, combined 0.135553.
- Accepted-epoch count 3, mean accepted code novelty 0.889336, and final complexity 0.58.
- Adaptation efficiency -0.003996 and archive sizes {'worst_cases': 5, 'failure_cases': 5, 'adversarial_layouts': 4} / elite 1.
- Panel heldout_cvrplib mean gap 0.239694 across 5 instances; family means: A=0.099476, B=0.241387, E=0.314779, P=0.301444.
- Panel synthetic_holdout mean gap 0.005376 across 4 instances; family means: alternating_belt=0.021505, clustered_demand=0.0, corridor_split=0.0, radial_heavy=0.0.
- Worst recent training cases: [{"best_known_cost": 458, "cost": 604, "family": "P", "name": "P-n40-k5", "optimality_gap": 0.318777}, {"best_known_cost": 937, "cost": 1065, "family": "A", "name": "A-n44-k6", "optimality_gap": 0.136606}, {"best_known_cost": 784, "cost": 860, "family": "A", "name": "A-n32-k5", "optimality_gap": 0.096939}]

cvrp_failure_replay

- Replay mode: failure with selection mode score_only.
- Final transfer gaps: CVRPLIB 0.237687, synthetic 0.005376, combined 0.134438.
- Accepted-epoch count 2, mean accepted code novelty 0.880282, and final complexity 0.58.
- Adaptation efficiency 0.061869 and archive sizes {'worst_cases': 5, 'failure_cases': 5, 'adversarial_layouts': 4} / elite 1.
- Panel heldout_cvrplib mean gap 0.237687 across 5 instances; family means: A=0.103839, B=0.247451, E=0.293666, P=0.296029.
- Panel synthetic_holdout mean gap 0.005376 across 4 instances; family means: alternating_belt=0.021505, clustered_demand=0.0, corridor_split=0.0, radial_heavy=0.0.
- Worst recent training cases: [{"best_known_cost": 458, "cost": 604, "family": "P", "name": "P-n40-k5", "optimality_gap": 0.318777}, {"best_known_cost": 937, "cost": 1073, "family": "A", "name": "A-n44-k6", "optimality_gap": 0.145144}, {"best_known_cost": 784, "cost": 860, "family": "A", "name": "A-n32-k5", "optimality_gap": 0.096939}]

cvrp_failure_replay_compression

- Replay mode: failure with selection mode novelty_gate.
- Final transfer gaps: CVRPLIB 0.198494, synthetic 0.0, combined 0.110274.

- Accepted-epoch count 2, mean accepted code novelty 0.766067, and final complexity 0.78.
- Adaptation efficiency 0.035528 and archive sizes {'worst_cases': 5, 'failure_cases': 5, 'adversarial_layouts': 4} / elite 2.
- Panel heldout_cvrplib mean gap 0.198494 across 5 instances; family means: A=0.075916, B=0.200286, E=0.238004, P=0.277978.
- Panel synthetic_holdout mean gap 0.0 across 4 instances; family means: alternating_belt=0.0, clustered_demand=0.0, corridor_split=0.0, radial_heavy=0.0.
- Worst recent training cases: [{"best_known_cost": 458, "cost": 611, "family": "P", "name": "P-n40-k5", "optimality_gap": 0.334061}, {"best_known_cost": 937, "cost": 1236, "family": "A", "name": "A-n44-k6", "optimality_gap": 0.319104}, {"best_known_cost": 672, "cost": 807, "family": "B", "name": "B-n31-k5", "optimality_gap": 0.200893}]

Judge Appendix

Summary of Results

Condition	Final CVRPLIB Gap	Synthetic Holdout Gap	Transfer Gap	Code Novelty (mean)	Replay Mode	Compression Pressure	Complexity	Adaptation Efficiency
cvrp_no_replay	0.244084	0.005376	0.137992	0.905812	none	No	0.58	0.03293
cvrp_random_replay	0.239694	0.005376	0.135553	0.889336	random	No	0.58	-0.003996
cvrp_failure_replay	0.237687	0.005376	0.134438	0.880282	failure	No	0.58	0.061869
cvrp_failure_replay_compression	**0.198494**	**0.0**	**0.110274**	**0.766067**	failure + compression	Yes	0.78	0.035528

Interpretation

- **Best transfer performance** (lowest **final CVRPLIB gap** and **synthetic holdout gap**, and smallest **transfer gap**) is achieved by **cvrp_failure_replay_compression**. This condition improves transfer gaps noticeably compared to others, with a CVRPLIB gap reduction from ~0.24 to ~0.20 (approx. 20% relative improvement) and synthetic gap improvement down to zero.

- **Compression-aware failure replay improves transfer significantly** over:
- no replay
- random replay
- failure replay without compression

- **Code novelty decreases** substantially (mean of 0.77 vs. ~0.88-0.91 in others) under compression-aware failure replay despite transfer improvements. This suggests:
- The transfer gains under compression are **not** due to increased lexical or superficial code novelty.

- Instead, the compression mechanism likely encourages a more stable or clustered constructor behavior (confirmed by the higher complexity 0.78 and "clustered_constructor" behavior profile).

- **No replay vs random replay vs failure replay (no compression):**
- Show minor differences in CVRPLIB gap (~0.237-0.244)
- Synthetic holdout gap is constant (0.005376)
- Transfer gap marginally decreases slightly from no replay (0.138) to failure replay (0.134)
- Code novelty slightly decreases from no replay to failure replay
- Adaptation efficiency highest for failure replay without compression (0.0619), but this does not translate into best transfer gap

- **Compression pressure increases model complexity** (0.78 vs 0.58). Yet, this complexity elevation coincides with improved transfer and zero synthetic gap, indicating more effective generalization.

Conclusion (Conservative)

- The **cvrp_failure_replay_compression** condition yields the strongest evidence of improved transfer ability, with reduced optimality gaps on held-out benchmarks and synthetic instances.
- This transfer improvement occurs *despite* a reduction in code novelty, indicating that lexical novelty alone does not drive transfer gains.
- Replay of failure cases combined with compression pressure promotes beneficial structural adjustments in the learned policy, improving generalization.
- Differences among no replay, random replay, and failure replay (no compression) are minor in terms of transfer and gap metrics.
- Claims of novelty or improvement should prioritize the demonstrated metrics of transfer and optimality gap rather than code novelty alone.

Notes on Replay Modes

Replay Mode	Description
none (cvrp_no_replay)	No experience replay
random (cvrp_random_replay)	Replay of random past experiences
failure (cvrp_failure_replay)	Replay focused on failure cases
failure + compression (cvrp_failure_replay_compression)	Replay of failure cases combined with compression pressure

Use of **failure replay with compression** most effectively leverages replay to improve transfer, outperforming other replay conditions.